CS230: Requirements Engineering
FSTE / SCIMS

Final Examination
Semester 1 2019

Mode *(Face -to -Face)*
Total Marks: 100

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 3 hours

Question / Answer Booklet

Instructions:

There are three sections in this paper, with a total of 10 questions

Section A 30 marks, Section B 40 marks, Section C 30 marks

This exam is worth 50 % of coursework

The minimum pass mark for this exam is 40 out of 100 marks

This booklet consists of 14 pages

This is a closed book Exam

Write the answers in this answer booklet

All the questions are compulsory. There are no choices.

Name: _____

ID: _____

Section A (Short Answers) 30 marks

1. a) Who are the stakeholders for a system? **(3 marks)**

Stakeholders are the people who have an interest or influence on the system, such as users, customers, sponsors, managers, regulators, or suppliers. They provide the requirements, expectations, feedback, and resources for the system. End users.

b) List five stakeholders for a healthcare patient information system. **(3 marks)**

Patients, Doctors, Receptionist, Vendors, Government, Policy makers

2. a) What is a requirement? **(2 marks)**

A rule the software must conform to: what it must do, how well, and within what constraints or limits.

b) How is a functional requirement different from a non-functional requirement? **(2 marks)**

Functional requirements define how the system must work and non functional requirements detail how it should perform. Without functional requirements the system will not work.

c) List the three types of non-functional requirements. **(3 marks)**

1. External
2. Product
3. Organizational

3. a) Differentiate Backward and Forward Traceability in Requirements management. **(3 marks)**

Tracing forward means using accumulated information to track the movement of products and tracing back means tracking records backward in the timeline.

b) Explain the term Change Management? **(2 marks)**

a systematic approach to dealing with the transition or transformation of an organization's goals, processes and technologies.

4. a) What is Program evolution dynamics? **(2 marks)**

the study of the processes of system change

b) Differentiate between Inception and Elaboration? **(1 mark)**

Elaboration focuses on requirements capture and determining the structure of the system. Inception is concerned with determining the scope and purpose of the project.

c) List and describe any two requirement elicitation techniques**. (3 marks).**

Questionnaire, interview, observation, background reading

5. Answer the questions that follow.

a) What four elements does Formal methods include. **(2 marks)**

Formal Specification
Formal Proof
Model Abstraction Checking
Translation of specifications

b) Explain the term Algebraic approach in Specification techniques? **(2 marks)**

formally defining types of data, and mathematical operations on those data types.

c) What two things are reviewed under requirements review? **(2 marks)**

The requirements review is an opportunity to check the requirements document for errors and omissions and confirm that the overall quality of the requirements is acceptable before the project proceeds. It is also one final opportunity to communicate and confirm project requirements with all

End of Section A

Section B (Long Answers) 40 marks

Refer to the following case study to answer questions 6.

Patient Record and Scheduling System

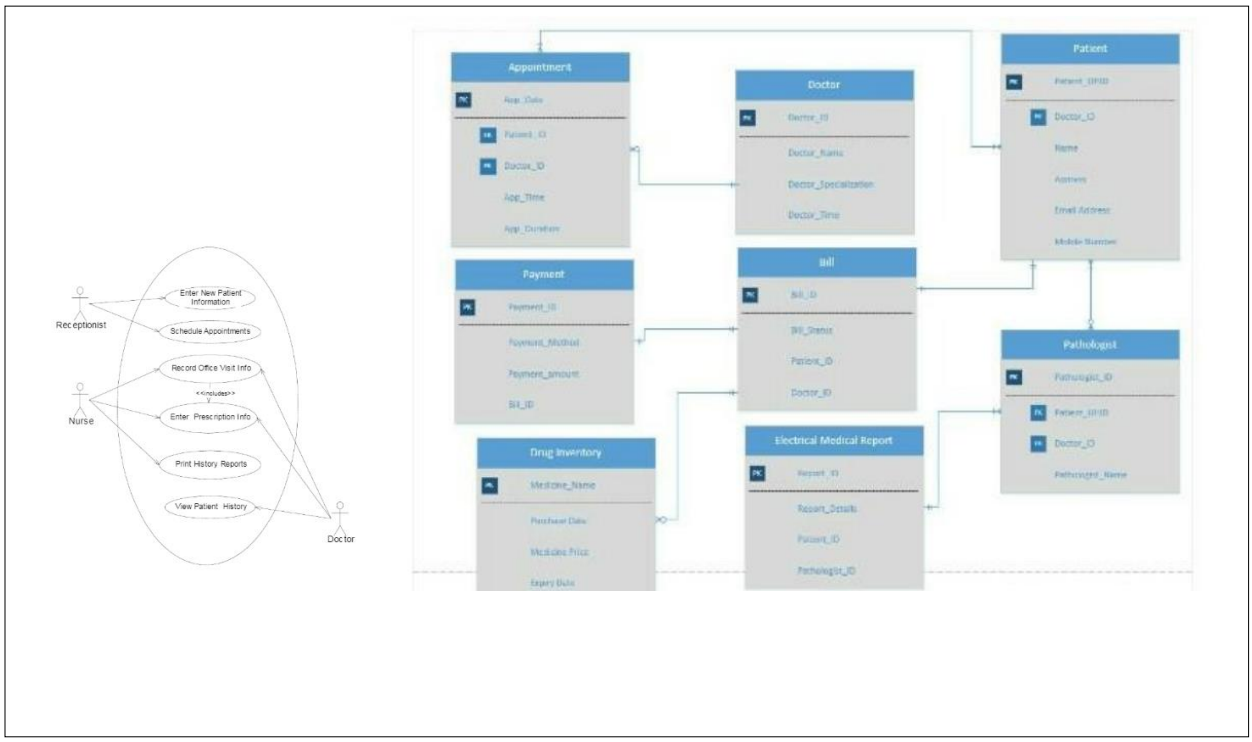
A patient record and scheduling system in a doctor's office is used by the receptionists, nurses, and doctors.

The receptionists use the system to enter new patient information when first-time patients visit the doctor. They also use the system to query available appointment slots and schedule all appointments. The nurses use the system to keep track of the results of each visit including diagnosis and medications. For each visit, free-form text fields are used to capture information on diagnosis and treatment. Multiple medications may be prescribed during each visit. The nurses can also access the information to print out a history of patient visits. The doctors primarily use the system to view patient history. The doctors may enter some patient treatment information and prescriptions occasionally, but most frequently they let the nurses enter this information.

For payment purposes, each patient is assigned to a family. The head of family is responsible for payment and is also the person with the primary medical coverage. Information about doctors is maintained since a family has a primary care physician, but different doctors may be the ones seeing the patient during the visit.

6. **UML: DFD and ERD** (10 Marks)

   a) Draw an ER diagram for the above scenario. (5 Marks)



b) Use the following description to create a Level-1 DFD for the process "Schedule Appointment", which will be used by the Receptionist to log an appointment. (Assume the appointment is made by a new patient) (5 marks)

Patients either visit or call the receptionist to request for an appointment. Once they are advised of the available time slots, they provide details such as name, address, date of birth, previous physician etc. The Receptionist feeds this information in the system. After the system validates all the entered information, it saves the new patient record in the patient file and records the appointment time. The receptionist then confirms the patient of the appointment time.

7. **Problem Domain and Modelling** (5 Marks)

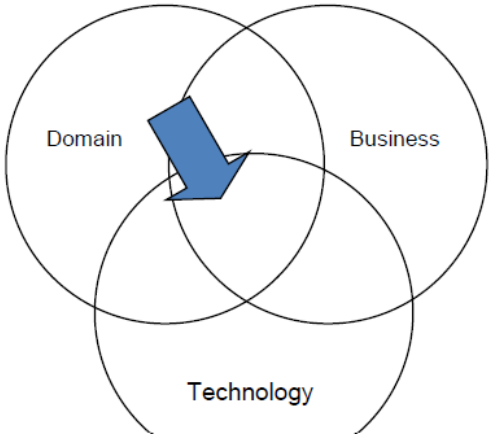
a) With a diagram illustrate the Domain-Specific Software Engineering. **(2 marks)**

- Domain
- Technology
- Business

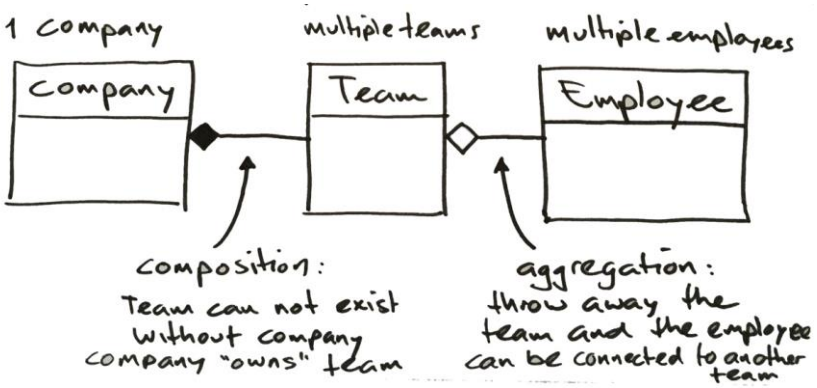
Domain + Business + Technology

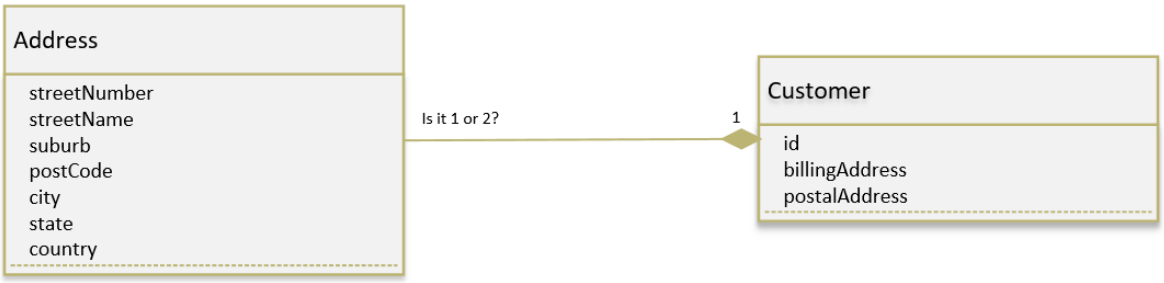
"Domain-specific software engineering"

Applies technology to domain-specific goals, tempered by business and market knowledge



b) Draw two classes (Country and City) and show association (multiplicity) relationship. **(3 marks)**





8. **Use Case Diagram and class diagram** (15 Marks)

Problem Description

Trade shows have become an important information source for new products, new fashions, and new fabrics. In addition to the large providers of outdoor clothing and fabrics, there are many smaller providers. It is important for RMO to capture information about these suppliers while the trade show is in progress. It is also important to obtain information about specific merchandise products that RMO plans to purchase.

Additionally, if quality photographs of the products can be obtained while at the trade show, then the creation of online product pages is greatly facilitated. It is recommended that a new system be developed and deployed so field purchasing agents can communicate more rapidly with the home office about suppliers and specific products of interest. This system should be deployed on portable equipment.

System Capabilities
The new system should be capable of:
• Collecting and storing information about the manufacturer/wholesaler (suppliers)
• Collecting and storing information about sales representatives and other key personnel for each supplier
• Collecting information about products
• Taking pictures of products (and/or uploading stock images of products)
• Functioning as a stand-alone without connection
• Connecting via Wi-Fi (Internet) and transmitting data
• Connecting via telephone and transmitting data

a) Identify and List the use cases with their **brief descriptions**. **(3 marks)**

1. **Capture Supplier Information**:
    - **Description**: Field agents collect and store details about manufacturers/wholesalers (suppliers) during the trade show.
    - **Actors**: Field purchasing agents, suppliers.
    - **Main Flow**:
        1. Agent enters supplier details (company name, contact information, address).
        2. System stores this information in the database.
    - **Alternate Flow**: If the supplier already exists, update the information.
2. **Record Sales Representatives' Information**:
    - **Description**: Gather and store data about sales representatives and other key personnel associated with each supplier.
    - **Actors**: Field agents, sales representatives.
    - **Main Flow**:
        1. Agent associates sales representatives with their respective suppliers.
        2. Record details (name, contact info, role) for each representative.
        3. Store this information in the system.
3. **Catalog Product Information**:
    - **Description**: Collect and manage data related to products available from suppliers.
    - **Actors**: Field agents, product managers.
    - **Main Flow**:
        1. Agent enters product details (name, description, category, price, etc.).
        2. Associate products with the corresponding suppliers.
        3. Store product information in the system.

b) Identify and list the objects/entities with their attributes for the system. **(2 marks)**

1. **Supplier**:
    - Attributes:
        - Supplier ID (unique identifier)
        - Company Name
        - Contact Information (phone, email)
        - Address
        - Website (if applicable)
2. **Sales Representative**:
    - Attributes:
        - Representative ID (unique identifier)
        - Full Name
        - Contact Information (phone, email)
        - Role (e.g., sales manager, account executive)
3. **Product**:
    - Attributes:
        - Product ID (unique identifier)
        - Name
        - Description
        - Category (e.g., clothing, fabrics)
        - Price
        - Supplier ID (foreign key linking to the supplier)

c) Draw a **Use-Case Diagram** representing the scenario given above. **(5 marks)**

d) Draw a Class diagram for the sytem. **(5 marks)**

9. **SRS** (10 Marks)

a) List and explain any two types of System Modelling? **(4 marks)**

1. Data model – structure pf the information
2. Behaviour model – How we interact with the system
3. Function & information flow model – what we will do with the data

b) Give two types of requirements that are not part of SRS. **(2 marks)**

Non-Functional Requirements and functional requirements

c) List and explain four Characteristics of a Good SRS. **(4 marks)**

1. Unambiguous
2. Complete
3. Verifiable
4. Consistent
5. Modifiable
6. Traceable
7. Usable during the Operation and Maintenance Phase

End of Section B

Section C (Case Study) 30 marks

10. Scenario based questions.

a) Feasibility report and Identifying and specifying software requirements (10 marks)

You are required to read the following system specifications and answer the questions below.

Tourist management system

Fiji and the pacific nations are known for its Tourist attractions. Be it the secluded heavenly beaches or the multi-cultural shrines or the native and exotic Fauna and Flora, Fiji attracts large number of tourist from all over the world. Tourists come from different part of the world and they may have used applications and tools that would assist them to visit places in their native countries. Hence, providing a similar tool for tourist would be useful to visitors and a boost to the tourism industry. Currently, there is no automated facility for visitors to access tourist information or plan their holidays. Most of the systems are primitive and are used by few hotels and travel agents that are tailor made to their own needs. You have been assigned by the Ministry of Tourism to develop a tourist portal (BULA Tourist) to assist visitors.

The portal should consist of the following modules:

1. *Destination Guide*
2. *Travel Planner*
3. *Eateries*
4. *Weather Guide*

The 'Destination Guide' module should help the visitors to view all the tourist locations in FIJI. The list of places should be supplemented with the complete history and details about the location, its proximity from the capital city, directions to drive or fly in and out of the locations. The module should use high quality images to project the exotic and divine nature of the locations.

The travel planner module should assist the visitors with all the necessary information to plan their travel to Fiji. It includes booking travel tickets, planning for accommodation and other essential travel bookings that you think would make the visitors comfortable. The module could have access to FIJI airways for travel bookings, if required. However your system would only interact with the FIJI airways and assist tourist with any travel bookings. Visitors should be provided with all the travel options while entering Fiji (including the information about Cruise ships and others) and commuting within the islands. Providing access to book local cabs would be highly appreciated.

The visitors should also be able to browse through the hotels that provide delicious dishes like the Fijian - Indian - Chinese food. The visitors should be able to make pre bookings if the hotels are busy during specific seasons of the year. You can use any existing tools that list all the eateries in Fiji or a similar module could be included in the portal.

The visitors should be well informed about the Climate in Fiji given its dynamic and ever changing weather. The portal could get access to the existing data from the Weather Bureau of Fiji. The stake holders namely the visitors, local tourist guides and travel agents should be able to access all the necessary information from the portal. You can identify unique fields while analysing each module and its requirements.

As there are numerous portals available elsewhere the portal is expected to match the standards of those in developed countries. Huge influx of tourists have been reported from neighbouring countries namely Australia and New Zealand and providing tailor made modules to attract them would not be a bad idea according to the tourist officials. Ensuring safety and security would be their utmost priority.

While preparing a feasibility report for the system specifications, you came across some sections such as Current Situation, Function and Non-Functional Requirements, etc. Based on the above scenario, provide the following in terms of a feasibility report:

i. Description of the current situation **(2 marks)**
ii. Functional requirements **(any four) 4 marks**
iii. Non-functional requirements **(any three) 3 marks**
iv. External interface requirements **(any one) 1 mark**

I. Description of the Current Situation:

The current situation in the tourism industry in Fiji lacks an efficient and automated system for visitors to access comprehensive tourist information and plan their holidays. Existing systems are primitive, fragmented, and tailored to specific hotels and travel agents. Visitors face challenges in obtaining accurate details about tourist locations, travel options, accommodations, and eateries. The absence of a centralized platform hinders the overall visitor experience.

II. Functional Requirements:

1. **Destination Guide Module**:
    - Display a list of all tourist locations in Fiji.
    - Provide detailed information about each location, including history, proximity to the capital city, and transportation options (driving or flying).
    - Showcase high-quality images to highlight the natural beauty of each destination.
2. **Travel Planner Module**:
    - Assist visitors in planning their travel:
        - Book travel tickets (including flights and cruise ships).
        - Plan accommodations (hotels, resorts, etc.).
        - Provide options for local commuting (local cabs, buses, etc.).
3. **Eateries Module**:
    - Allow visitors to browse through restaurants and eateries in Fiji.
    - Display information about Fijian, Indian, and Chinese cuisine.
    - Enable pre-bookings for busy seasons.
4. **Weather Guide Module**:
    - Access real-time weather data from the Weather Bureau of Fiji.
    - Provide climate information to help visitors prepare for their trip.
    - Ensure safety by alerting visitors about extreme weather conditions.

III. Non-Functional Requirements:

1. **Performance**:
   - The portal should respond quickly to user requests, especially during peak tourist seasons.
   - Load times for images and data should be optimized.
2. **Security**:
   - Ensure data privacy and protection of personal information.
   - Implement secure authentication for user accounts.
3. **Usability**:
   - The user interface should be intuitive and user-friendly.
   - Visitors of all ages and backgrounds should find it easy to navigate.

IV. External Interface Requirements:

- **Integration with FIJI Airways**:
  - The system should interact with FIJI Airways for travel bookings.
  - Provide seamless access to flight information and booking services.

b) Read the problem statement given below very carefully and use it to answer the questions that follow. (20 marks)

SCIMS Lab Booking System

Each semester SCIMS staff planning to use the ICT Center Teaching Labs would send their lab booking request to the SCIMS IT personnel via email. For example, in CS240 I had reserved the four one-hour lab sessions by sending an email to our IT personnel. You are required to design a web based Booking System that will allow staff to book their respective lab classes.

The SCIMS IT personnel (referred to as the IT Admin) will be allowed to enter the following data about each computer lab: Lab ID (unique for each lab), Description (used for a brief description of the lab), and Number of computers in the lab. The Booking System should allow the lab details to be changed by the IT Admin. The IT Admin will also be responsible for adding or updating users of the system. For each user/staff the following data would be stored: Staff ID (unique), username, password, name of the staff, description (for any additional data (optional)). The IT Admin will also add/update/delete the list of courses offered by SCIMS. The data recorded for each course includes the course code, and title.

A staff intending to book a lab will have to login first. After logging into the Booking System, the staff can book available labs. When booking a lab class, the teaching staff is required to provide the following data: Lab ID of the lab to be used, Course Code (e.g. CS230) of the course that the lab will be utilized, day

(Monday – Friday), start time, and end time of the lab class. When this data is submitted to the system, the system will book the particular lab on the given day and time. The system should also store the Staff ID of the staff who is booking the lab and send a notification to the IT Admin for approval. The IT Admin needs to approve each booking request, and after each approval an email is sent to the staff that did the booking informing him/her about the decision of the IT Admin. The IT Admin can at any time cancel/delete a particular confirmed booking and an email will be sent to the staff that made the booking informing the staff of this case. Note that the teaching staff who booked a particular lab class can change or cancel their booking as well. The IT Admin can view a report on all cancelled lab classes. We will assume that the labs operate from 9am to 8pm from Monday to Saturday. The IT Admin will also print a report showing the number of lab classes for each course in a particular semester.

All users, including IT Admin should be able to change their profile, which includes, name and password. The Booking System is also required to send an email once every week to the IT Admin informing him about the total number of new bookings done. The homepage of the Booking System should display the current timetable of the labs. You should also include a page that shows the teaching staff how to book a lab class.

i. Create a detailed use case diagram for the SCIMS Lab Booking System capturing the functional requirements. (10 marks)

ii. One of the events of the SCIMS Lab Booking System is **SCIMS staff booking a computer lab** for their course's lab classes. The following description provides an idea of how this can be achieved.

The Booking system displays the courses offered in the current semester and year, and the current lab timetable. The SCIMS staff starts by choosing the course. All the lab classes that the staff wants will be entered into the system. This means that a staff can book many lab classes at once. After this is done, the staff will submit the booking request, and the system records these details. A notification is also sent to the IT Admin for approval.

Using the above description, write a **use case scenario** showing the Brief Description, Actors and the Flow of Events (FoE)/Activities. For the FoE show the Actor and System tasks. (6 marks)

Brief Description:

The use case involves SCIMS staff booking computer labs for their course's lab classes. The system displays available courses, lab timetables, and allows staff members to select and book multiple lab classes. After submission, the system records the booking details and notifies the IT Admin for approval.

Actors:

- **SCIMS Staff**: The staff members responsible for booking lab classes.
- **IT Admin**: The administrator who approves or manages lab bookings.

Flow of Events (FoE):

1. **SCIMS Staff Initiates Booking**:
   - **Actor**: SCIMS Staff
   - **System Task**:
     1. Staff accesses the lab booking system.
     2. Views available courses for the current semester and year.
     3. Chooses the relevant course.
2. **Selecting Lab Classes**:
   - **Actor**: SCIMS Staff
   - **System Task**:
     1. Staff views the lab timetable for the selected course.
     2. Identifies the lab classes they want to book.
     3. Enters details of the desired lab classes (date, time, lab room).
3. **Submitting Booking Request**:
   - **Actor**: SCIMS Staff
   - **System Task**:
     1. Staff submits the booking request.
     2. The system records the booking details (course, lab classes).
     3. Generates a notification for the IT Admin.
4. **IT Admin Approval**:
   - **Actor**: IT Admin
   - **System Task**:
     1. IT Admin receives the booking notification.
     2. Reviews the request.
     3. Approves or rejects the booking.
     4. Sends confirmation/notification to the SCIMS Staff.

iii. Now create a System Sequence diagram for the scenario you have given in part (ii) above. (4 marks)

THE END